Define the row before joining

The source record is wide. The two warehouse facts express different analytical grains.

Table	Type and grain
FactHourlyDemand	Hourly Periodic Snapshot Fact. One selected authority and one UTC hour-ending instant with its demand, forecast and total net generation.
FactHourlyGeneration	Hourly Periodic Snapshot Fact. One selected authority, UTC hour-ending instant and original fuel category with a numeric reported value.

Structure	Business identity
DimDate	CalendarDate
DimHour	HourEnding
DimBalancingAuthority	AuthorityCode
DimEnergySource	SourceColumn
DimSourceSnapshot	(FileName, SHA256)
FactHourlyDemand	(AuthorityKey, UTCHourEnd)
FactHourlyGeneration	(AuthorityKey, UTCHourEnd, EnergySourceKey)

One real source example

CISO, operating date 1 January 2025, local end 01:00 and UTC end 09:00. Reported demand is 22,160 MW, forecast is 21,524 MW and total generation is 15,115 MW. This creates one demand row. Each numeric reported fuel cell creates a separate generation row with its original category key.

That hour includes natural gas of 8,666 MW and nuclear of 2,274 MW. Joining raw fuel rows to the demand row repeats 22,160 for every source category. First aggregate the fuel population to one authority-hour result, then join it to demand once.

Time, missing values and revisions

Different local labels and different source categories do not change the authority-hour identity.

Daylight-saving time

Parse UTC Time at End of Hour with an explicit UTC offset. Parse Local Time at End of Hour as the source wall-clock label. Store their difference as UTCOffsetMinutes. The two autumn repetitions per authority remain distinct because their UTC instants differ. A midnight end label can belong to the previous operating date, so DateKey comes from Data Date.

Sparse generation observations

A valid zero generation cell creates a row. A blank cell creates no numeric generation fact. Distinguish a missing reported value from a category that did not exist in that source schema. The generation count is an observation count, never a count of operating hours.

Full versus incremental loads

For the first implementation load, process the four frozen baseline files. Later, re-fetch the current six-month file into a new extraction folder and compare its hash. If changed, process the changed partition by business key. Recheck older files periodically for corrections. The script supports selecting a year/half and refreshing downloads; warehouse upserts and load watermarks are Phase 2 work.

Revision handling

Keep immutable raw extracts and their hashes. A changed bulk file creates a new DimSourceSnapshot member. A later load updates measures at the same business grain and preserves fact surrogate IDs. Replacement of a source partition must also remove fuel facts for values withdrawn by the publisher, or mark them absent through a reviewed policy. A latest-value fact cannot reproduce an old publication without its archived source snapshot.

Join validation

A correct join preserves the demand count and total energy for the selected authority-hour cohort. For Q8/Q9, filter both populations to the same matched hours and recalculate all denominators after filtering.